Chapter 3: Introduction to Web Applications

Our Python database script carried out a sequence of operations and stopped. A web application waits for requests. Someone opens a page, submits a form, or follows a link, and the application decides what work that request requires.

The example in `topic-03-intro-to-web-apps` puts a browser interface on the pet database. Python still issues Structured Query Language (SQL) statements through `sqlite3`. What changes is how we choose an operation and where its inputs come from. The application reads a request, works with stored state, and sends back a response.

This chapter follows `app.py`, the files in `templates`, and the address examples in `example.txt`. Code excerpts retain the example's behavior, with formatting adjusted for readability. Additional examples and suggested changes are identified separately.

Running the Application

Work from the `topic-03-intro-to-web-apps` directory. It contains `app.py`, the supplied `pets.db`, and the `templates` folder. Make a working copy of this directory for practice; creating, editing, and deleting pets changes its database.

For a new Python environment on macOS or Linux:

```
python3 -m venv .venv
source .venv/bin/activate
pip install flask
flask run
```

Flask discovers the application in `app.py`. The explicit equivalent is `flask --app app run`. The terminal shows the address to open, normally:

```
http://127.0.0.1:5000/
```

Open `/pets` at that address to see the database interface. Leave the terminal running while using the application. Stop the server with Control-C. If port 5000 is already occupied, use `flask run --port 5001` and open the address it reports. This is Flask's development server, intended for local development.

Unlike the SQL-in-Python demonstration, this application does not rebuild the pet table at startup. It uses the existing database. Running from the wrong directory can create a new, empty `pets.db`; opening a connection successfully does not mean that file contains the expected table.

What an Address Tells Us

A Uniform Resource Locator (URL) identifies the resource a browser is requesting. The topic's `example.txt` takes an address apart. Here is a similar example with the parts together:

```
https://mywebsite.com:8443/catalog?category=blenders&brand=KitchenAid
```

Part	Meaning
`https`	Scheme: use encrypted Hypertext Transfer Protocol (HTTP)
`mywebsite.com`	Hostname
`8443`	Explicit port number in this example
`/catalog`	Path requested from the server
`category=blenders&brand=KitchenAid`	Query string containing named values

The Domain Name System (DNS) resolves a hostname to an Internet Protocol (IP) address. A port helps direct the connection to a service at that address. When omitted, the usual port is 443 for HTTPS and 80 for HTTP. Our local Flask server uses HTTP on port 5000 instead.

A path is not necessarily a file on the server. `/pets` selects application code that builds a page. There is no requirement for a file named `pets` to exist.

The notes also contrast values in a path with values in a query string:

```
/catalog/blenders
/catalog?category=blenders
```

Either could represent a category selection, but the application must implement the chosen convention. Flask does not infer a database search from the word `blenders`. These catalog addresses are discussion examples; the pet application does not implement them. Query values do not need literal quotation marks around them.

Requests, Routes, and Responses

A browser sends an HTTP request containing a method and a target address. A `GET` requests a representation, such as a page or an edit form. In this application, `POST` submits form values for creating or updating a record.

Flask matches the request to a route and calls its Python function. The function's return value becomes a response. A response includes a status code and may include a body, such as Hypertext Markup Language (HTML), or a redirect telling the browser to request another address.

The application starts with:

```
from flask import Flask, render_template, request, redirect, url_for

import sqlite3
from pprint import pprint

connection = sqlite3.connect("pets.db", check_same_thread=False)
print("succeeded in making connection.")

app = Flask(__name__)
```

`app` is the Flask application object. The connection is an in-memory Python object through which the application accesses the database file. `pprint` prints debugging output in the server terminal; it does not put anything on the browser page.

The simplest route is:

```
@app.route("/bye", methods=["GET"])
def get_bye():
    return "Bye!"
```

The decorator registers the function for `GET /bye`. Defining the function does not send a page to anyone. Flask calls it when a matching request arrives.

Here are the main operations in the pet interface:

Request	Function	Work
`GET /pets`	`get_pets`	Load records and display the table
`GET /create`	`get_create`	Display a blank form
`POST /create`	`post_create`	Insert a pet, commit, and redirect
`GET /update/<id>`	`get_update`	Load a pet and fill an edit form
`POST /update/<id>`	`post_update`	Save submitted changes and redirect
`GET /delete/<id>`	`get_delete`	Delete a pet and redirect

The deletion route is a shortcut in this teaching example. A `GET` should not change stored data. We will return to that distinction when examining the delete function.

Passing Values to a Template

The greeting function has three routes:

```
@app.route("/", methods=["GET"])
@app.route("/hello", methods=["GET"])
@app.route("/hello/<name>", methods=["GET"])
def get_hello(name="world"):
    return render_template("hello.html", name=name)
```

`/` and `/hello` use the default name, `world`. A request for `/hello/Ada` supplies the string `"Ada"` as the function argument. The `<name>` part is a variable in the route, not text the user types literally.

The template `hello.html` contains:

```
<html>
    <body>Hello, {{ name }}! (from the template)</body>
</html>
```

`render_template` evaluates this file using Jinja, Flask's template engine. The keyword argument `name=name` makes the Python value available under the template name `name`. The browser receives HTML containing the completed greeting. It does not run the Python function or evaluate the Jinja expression.

The double braces insert a value. In these HTML templates, Flask enables automatic escaping, so characters such as `<` in an ordinary supplied string are displayed as text rather than interpreted as an HTML tag.

This is our first separation of responsibilities: the route decides what data to provide, and the template describes the resulting page.

Loading and Displaying Pets

The list route reads the database before rendering its page:

```
@app.route("/pets", methods=["GET"])
def get_pets():
    cursor = connection.execute("select * from pet")
    rows = cursor.fetchall()
    pprint(rows)
    return render_template("pets.html", pets=rows)
```

The supplied table has columns `id`, `name`, `kind`, `age`, and `food`, in that order. Each result row is a tuple. `rows` is a list of those tuples, passed to the template as `pets`.

The query has no `ORDER BY`, so the application should not depend on a particular row order. An explicit ordering would be needed to promise an alphabetical display, for example.

Inside the table in `pets.html`, the template uses nested loops:

```
{% for pet in pets %}
<tr>
    {% for item in pet %}
    <td>{{ item }}</td>
    {% endfor %}
    <td><a href="/update/{{pet[0]}}">
        <span class="material-symbols-outlined">edit</span>
    </a></td>
    <td><a href="/delete/{{pet[0]}}">
        <span class="material-symbols-outlined">delete</span>
    </a></td>
</tr>
{% endfor %}
```

`{% ... %}` marks template statements, including loops. The outer loop creates a table row for each pet. The inner loop creates a cell for each column value.

`pet[0]` is the record's identifier. It becomes part of the edit and delete links. For a pet whose identifier is `3`, the edit link points to `/update/3`. A name might change or be shared by several pets; the identifier selects the stored record.

The table's appearance uses Bootstrap styles and externally loaded icon fonts. Those resources affect presentation. The database query and template loops still determine the actual records and links. The separate `list.html` file belongs to a commented-out list demonstration; the current route renders `pets.html`.

Displaying a Form Is Not Submitting It

The create page begins with a `GET`:

```
@app.route("/create", methods=["GET"])
def get_create():
    return render_template("create.html")
```

The form in that template is:

```
<form action="/create" method="post">
    <p>Pet Name:<input name="name"/></p>
    <p>Kind:<input name="kind"/></p>
    <p>Age:<input name="age"/></p>
    <p>Food:<input name="food"/></p>
    <button type="submit">Create</button>
    <a href="/pets">Cancel</a>
</form>
```

The `action` specifies where to submit the form. The `method` specifies how. Each input's `name` becomes the key associated with its submitted value. The text `Pet Name:` is a label for the reader; the application receives the key `name` because of the input attribute.

Opening this form does not insert a pet. Typing into it does not insert one either. Clicking Create sends a separate `POST /create` request with the form values in the request body. Cancel is an ordinary link back to `/pets` and does not submit the form.

Creating and Saving a Pet

The `POST` route handles the submitted values:

```
@app.route("/create", methods=["POST"])
def post_create():
    data = dict(request.form)
    cursor = connection.execute(
        """insert into pet(name, kind, age, food)
           values (?,?,?,?)""",
        (data["name"], data["kind"], data["age"], data["food"])
    )
    rows = cursor.fetchall()
    connection.commit()
    return redirect(url_for("get_pets"))
```

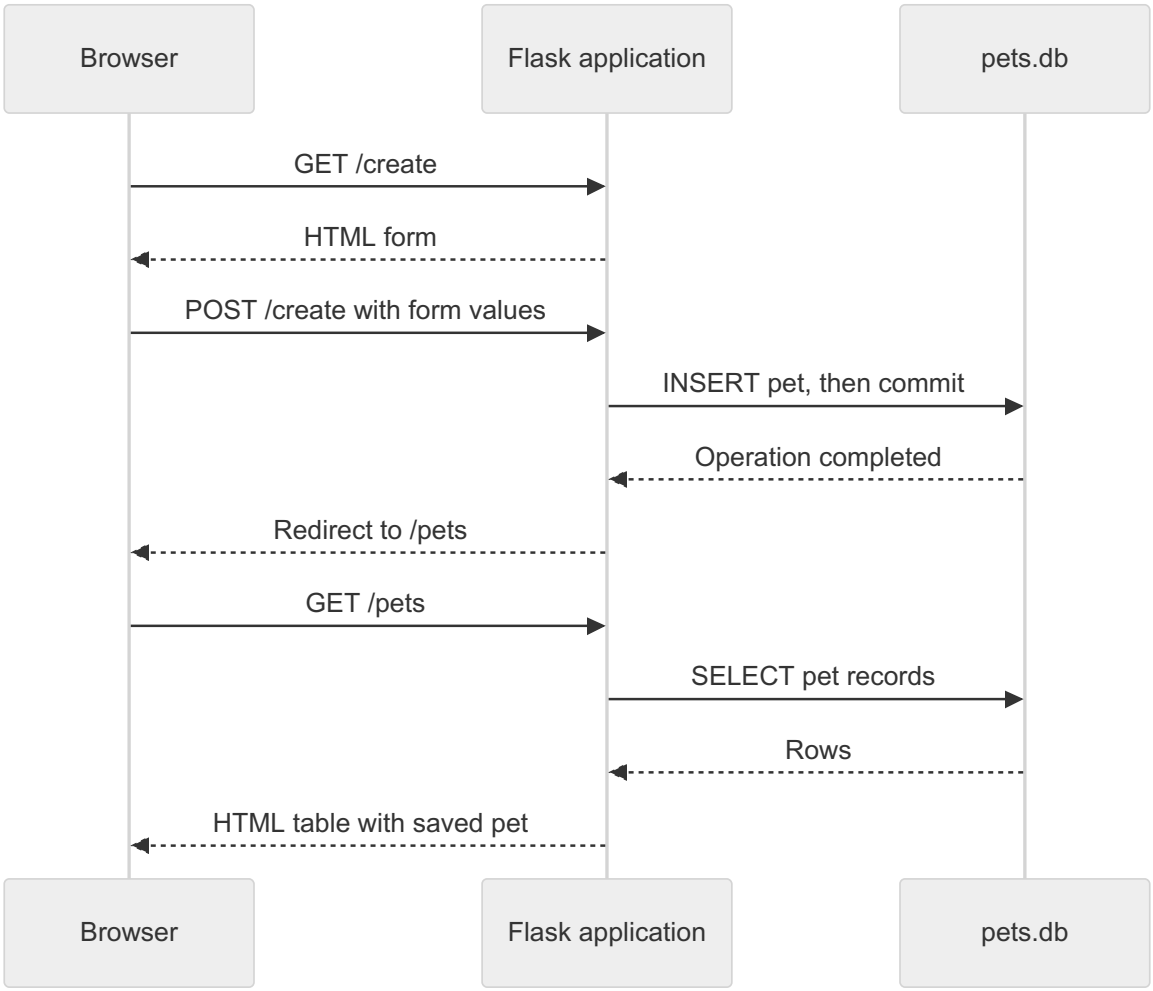
`request.form` contains parsed form data for this request. Converting it to a dictionary gives the function a familiar way to retrieve the fields. These input values arrive as strings, including age.

The SQL statement leaves out `id`; SQLite supplies the identifier for the new record. The question marks bind the submitted values as data. A pet named `O'Malley` does not require the application to build an SQL string with an escaped apostrophe.

`fetchall()` is unnecessary here. This `INSERT` has no `RETURNING` clause and produces no result rows. The statement has already executed; fetching is not what makes the insertion happen. `commit()` completes the transaction so the change is saved.

Finally, `url_for("get_pets")` finds the URL associated with that function name. `redirect()` returns a redirect response, normally with status `302`. The browser follows it with `GET /pets`, which queries the database again and renders the updated list.

A Create Request, Followed by a Fresh Display



The form submission and the final display are separate requests. Refreshing the final list requests that list again, rather than ordinarily resubmitting the insertion. This pattern is often called Post/Redirect/Get.

Loading a Record for Editing

Editing begins by loading the selected pet. The route `/update/<id>` supplies an identifier as text; `get_update` converts it with `int(id)` before querying:

```
cursor = connection.cursor()
cursor.execute(f"""select * from pet where id = ?""", (id,))
rows = cursor.fetchall()
```

The `f` prefix on this string is unnecessary: there is no Python expression inside it. The value is supplied through the SQL parameter, `(id,)`, whose trailing comma makes it a one-element tuple.

For a matching record, the function unpacks the row and constructs a dictionary:

```
(id, name, kind, age, food) = rows[0]
data = {
    "id": id,
    "name": name,
    "kind": kind,
    "age": age,
    "food": food
}
```

It then returns `render_template("update.html", data=data)`. This dictionary represents the loaded record in application memory. Creating or changing the dictionary does not change the database.

The edit template uses the values to fill the form. These lines are excerpts:

```
<form action="/update/{{data['id']}}" method="post">
    <p>Pet Name:<input name="name" value="{{data['name']}}"/></p>
    <p>Kind:<input name="kind" value="{{data['kind']}}"/></p>
    <p>Age:<input name="age" value="{{data['age']}}"/></p>
    <p>Food:<input name="food" value="{{data['food']}}"/></p>
```

For record 3, the completed form submits to `/update/3`. The `value` attributes provide the starting contents of the inputs. Once the page reaches the browser, the person can edit those inputs without changing the server's earlier dictionary or the stored row.

Submitting an Edit

The save operation uses `POST /update/<id>`. It obtains the submitted fields and converts age:

```
data = dict(request.form)
try:
    data["age"] = int(data["age"])
except:
    data["age"] = 0
```

This is the example's fallback behavior, not thorough validation. An entry such as `unknown` silently becomes zero. A more useful response would explain the problem and let the person correct it before saving.

The update itself is parameterized:

```
cursor = connection.cursor()
cursor.execute(
    """update pet set name=?, kind=?, age=?, food=? where id=?""",
    (data["name"], data["kind"], data["age"], data["food"], id)
)
rows = cursor.fetchall()
connection.commit()
return redirect(url_for("get_pets"))
```

The identifier chooses the row; the other parameters supply replacement values. Again, `fetchall()` has no useful result to fetch. After committing, the redirect leads to a fresh query and display.

This is the persistence cycle spread across browser requests. The edit link determines the action and selects a record. The first request loads its state. The person changes the form, and a later request saves those changes. Display happens both when the form is filled and when the updated list is shown.

HTTP does not automatically connect those requests into one continuing Python function call. The identifier in the form action tells the save route which record to update. The database preserves the state between requests and after the application stops.

Deleting and Handling Missing Records

Deletion is short:

```
@app.route("/delete/<id>", methods=["GET"])
def get_delete(id):
    id = int(id)
    cursor = connection.execute(
        """delete from pet where id==?""", (id,)
    )
    connection.commit()
    return redirect(url_for("get_pets"))
```

SQLite accepts `==` as an equality operator. The database operation is familiar, but attaching it to a `GET` link is a poor convention for a finished application. Merely following a link should not delete a record. A revised interface should submit a `POST` for deletion and provide protection against cross-site request forgery, where another site induces an unwanted request.

The edit route also demonstrates an error template. With no identifier, it displays `No ID was provided.` With no matching row, accessing `rows[0]` fails and the function displays `Data not found.`

Displaying an error message is separate from choosing an HTTP error status. These template responses still have the default status `200`. Also, a nonnumeric identifier fails at `int(id)` before the function's existing exception handler. Those are useful distinctions when testing: invalid input, an absent record, and a successful request should not be treated as the same outcome.

There is a small routing detail in the source: both update functions have an extra `@app.route("/update")` decorator. Without a `methods` argument, it registers a `GET`, even on the function named `post_update`. The working save route is `POST /update/<id>`; `POST /update` is not registered. Function names help us read the code, but the decorators determine which requests Flask accepts.

What Persists, and What Needs More Work

The browser has form fields. Python has strings, dictionaries, and result tuples. SQLite has stored records. Values move between these representations, but they do not remain automatically synchronized. Every save in this application is an explicit database operation.

The example keeps a single database connection at module level. `check_same_thread=False` disables SQLite's Python thread check; it does not make a shared connection's concurrent operations safe. A connection with a defined request lifetime is a better direction as the application grows. Flask's database tutorial in the readings shows that approach.

For now, trace one operation all the way through. Identify the request, find its route, follow the database work, and inspect the response. That makes it much easier to tell whether a problem is in the form, the Python code, or the stored data. The database-abstraction topic will give us a place to collect database operations instead of repeating them throughout route functions.

Practice

Use a working copy of the topic directory and its database.

1. Start the application with `flask run`. Visit `/`, `/hello/Ada`, `/bye`, and `/pets`. Match each response to its route function and template, if any.
2. Create a pet named `O'Malley`. Find the record using the SQLite shell. Compare the submitted form fields with the stored columns, including the generated identifier.
3. Open the browser's developer tools and inspect network requests while creating a pet. Find the form submission, redirect, and subsequent list request. Record their methods and status codes.
4. Open an edit form and change a field without submitting it. Query the database from the shell. Then submit the form and query again. Explain when the stored value changes.
5. Change the list query to display pets in name order. Keep the template unchanged and verify the result.
6. In your copy, make invalid ages produce an error message instead of saving zero. Verify that an invalid submission leaves the stored record unchanged.
7. Replace the delete link with a small form that submits a `POST`. Change the route's allowed method to match. Verify that a direct `GET` no longer deletes a pet.
8. Create a pet, stop the server, and start it again. Confirm that the record remains. Explain why this differs from restarting the earlier script that rebuilt its table.

Questions for Thought and Study

1. Why can `/pets` work when there is no file named `pets`?
2. How can `GET /create` and `POST /create` call different functions?
3. Which parts of `hello.html` are evaluated by Jinja, and what does the browser receive?
4. What connects a submitted form field to `data["name"]`?
5. Why does an input named `age` arrive as a string?
6. What is the difference between a record identifier and its position in the displayed table?
7. Why does changing the edit form not immediately change the database?
8. What does the redirect accomplish after a successful insert?
9. Why should deleting a record require more than following a `GET` link?
10. Which state survives a server restart, and which objects must be created again?

Further Reading and Practice

- **Web applications:** Background on applications accessed through a browser.
  https://en.wikipedia.org/wiki/Web_application
- **Flask quickstart:** Running the server, routes, templates, and request data.
  https://flask.palletsprojects.com/en/stable/quickstart/
- **Flask database tutorial:** Managing a database connection during a request.
  https://flask.palletsprojects.com/en/stable/tutorial/database/
- **Sending form data:** How form actions, methods, and named controls determine a request.
  https://developer.mozilla.org/en-US/docs/Learn_web_development/Extensions/Forms/Sending_and_retrieving_form_data